American Computer Science League

2025-2026 ● Contest 3: Short Problems ● Classroom Division

1. Boolean Algebra

Which ordered triple makes the following Boolean expression FALSE?

$$A(\overline{B + C}) + \overline{ABC}$$

A. (1,0,1)
B. (0,0,0)
C. (1,1,0)
D. (1,1,1)
E. (0,0,1)

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{AB}\ (\overline{A + C}) + \overline{C}$$

A. $\overline{A}$
B. $\overline{B}$
C. $\overline{C}$
D. $\overline{A} * \overline{C}$
E. $\overline{A} * \overline{B} * \overline{C}$

3. Data Structures

Build a binary search tree for the string

MASSACHUSETTS

How many nodes have exactly one child?

A. 3
B. 4
C. 5
D. 6
E. 9

4. Data Structures

Given an initially empty stack and the following commands on the stack, what will be popped next?

```
PUSH("B")
PUSH("O")
x = POP()
PUSH("S")
x = POP()
PUSH("T")
PUSH("O")
x = POP()
x = POP()
PUSH("N")
x = POP()
```

A. S
B. T
C. O
D. N
E. B

5. Boolean Algebra

Simplify the following Boolean algebra expression:

$$A\overline{(B + C)} + \overline{B\overline{C}} + C$$

A. $B + C$
B. $\overline{B} + C$
C. $B + \overline{C}$
D. $\overline{B} + \overline{C}$
E. $B\overline{C}$

6. Boolean Algebra

How many ordered triples make the following Boolean algebra expression TRUE?

$$\overline{A\overline{(B + C)} + \overline{B + \overline{C}}}$$

A. 3
B. 5
C. 6
D. 7
E. 8

7. Data Structures

Build a binary search tree for the string

CALIFORNIA

What is the internal path length?

A. 17
B. 19
C. 20
D. 21
E. 23

8. Data Structures

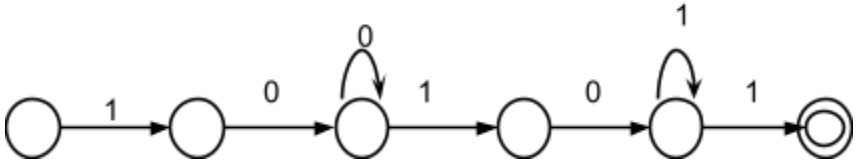
Given an initially empty queue and the following commands on the queue, what item will be popped next?

```
PUSH("S")
PUSH("A")
PUSH("C")
x = POP()
PUSH("R")
PUSH("A")
x = POP()
x = POP()
PUSH("M")
PUSH("E")
x = POP()
PUSH("N")
PUSH("T")
PUSH("O")
x = POP()
```

A. N
B. T
C. O
D. E
E. M

9. FSAs and Regular Expressions

What is the regular expression for the following FSA below?



A. 1001011
B. (10101)*
C. 10*101*
D. 10*0011*1
E. 100*101*1

10. FSAs and Regular Expressions

How many of the following words are accepted by this regular expression?

```
[a-ct-v][e-m|o-u]*[aeiou]?
```

violin	cello	trombone
trumpet	oboe	saxophone
tuba	flute	clarinet

A. 1
B. 2
C. 3
D. 4
E. 5